Agglomerative Clustering - Explanation with Working Example

Agglomerative Clustering is a hierarchical clustering algorithm used in unsupervised machine learning. It works by treating each data point as an individual cluster initially and then repeatedly merging the closest clusters until only one large cluster remains.

How Agglomerative Clustering Works

1. Start with each data point as its own cluster.
2. Calculate the distance between all clusters.
3. Merge the two closest clusters.
4. Recalculate distances between new clusters.
5. Repeat until all points belong to one cluster or the desired number of clusters is reached.

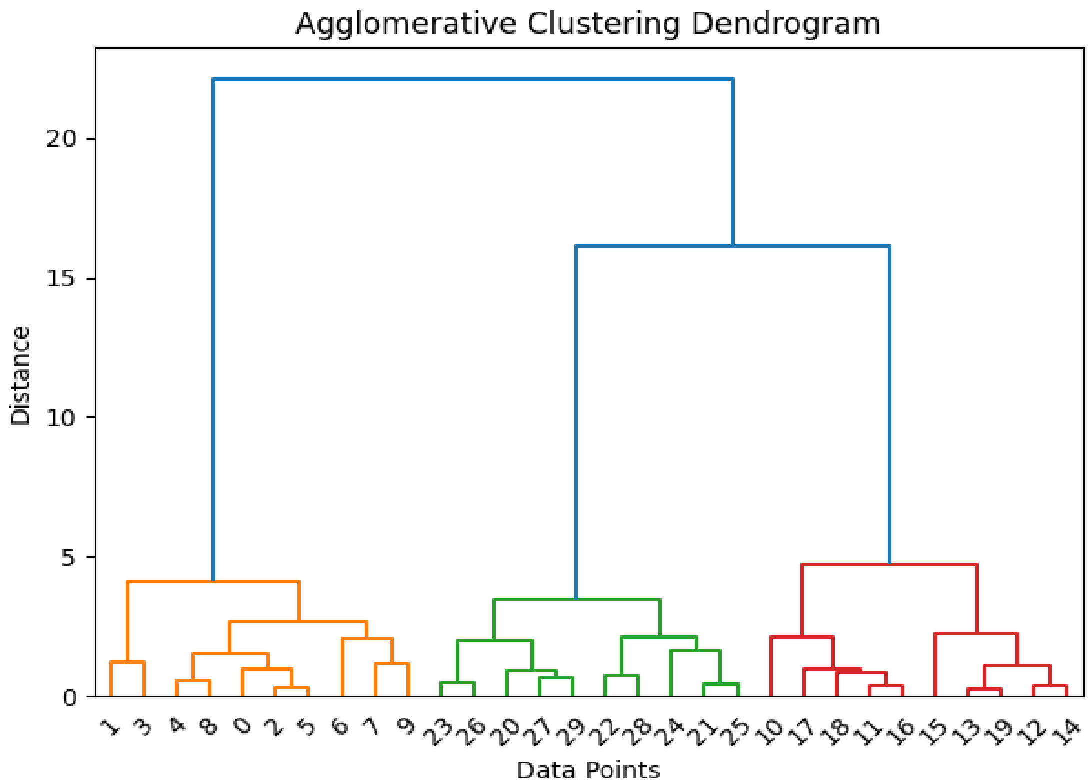
Distance Linkage Methods

Agglomerative clustering can use different linkage methods: **Single Linkage:** Minimum distance between clusters. **Complete Linkage:** Maximum distance between clusters. **Average Linkage:** Average distance between clusters. **Ward Linkage:** Minimizes variance within clusters.

Working Example

Suppose we have customer purchasing data and want to group similar customers.

Initially: Each customer is treated as a separate cluster. The algorithm calculates distances between customers. Closest customers are merged step-by-step. A dendrogram is formed showing the hierarchy of clusters. The final clusters can help businesses understand customer behavior.



Advantages of Agglomerative Clustering

No need to specify the number of clusters initially. Produces a hierarchical structure (dendrogram). Useful for small datasets and exploratory analysis.

Limitations of Agglomerative Clustering

Computationally expensive for large datasets. Once clusters are merged, they cannot be split again. Sensitive to noise and outliers.

Conclusion

Agglomerative Clustering is a powerful hierarchical clustering method that builds clusters gradually. Its dendrogram visualization helps in understanding relationships between data points and choosing the optimal number of clusters.